Western Division American Fisheries Society Conference
Portland, OR
Survey Design Workshop
Presented by Western EcoSystems Technology, Inc.
May 4, 2026

SAMPLING TERMS

- **Target population**: The complete collection of units to which inference is made
- **Sampling frame**: The list of sampling units. In the example, the sampling frame would be the list of all reaches in the watershed.
- **Sample**: A subset of a population. This may be chosen through a random, probabilistic mechanism (random sampling) or nonrandomly (judgment sampling)
- **Sampling unit**: The unit actually sampled. For example, we may want to survey all redds in the four basins, but we do not have a list of all redds by basin. Reaches are used as the sampling units.
- **Sampled population**: The collection of all possible sampling units that might have been chosen from a sample. This is the population from which the sample was actually taken.
- **Response unit** (or observation unit): The object on which the measurement is taken. This is the basic unit of observation. Redds will be the response/observation units within selected reaches.

NOTATION

Y = outcome of interest

τ = population total

μ = population mean

N = population size (finite domain)

$|R|$ = population size (continuous domain)

n = sample size

n' = number of target units in the sample

m = number of responding units in the sample

π_i = sampling inclusion probability for sampling unit *i*

w_i = sampling design weight for sampling *i*; the extent of the sampling frame represented by sampling unit *i* in the sample

EQUATIONS

Inclusion probability for a SRS: $\pi_i = \frac{n}{N}$

Design weight for unit *i*: $w_i = \frac{1}{\pi_i}$

Horvitz-Thompson Estimator:

$$\hat{\tau} = \sum_{i=1}^{N} \frac{Z_i y_i}{\pi_i} = \sum_{i=1}^{n} \frac{y_i}{\pi_i} = \sum_{i=1}^{n} w_i y_i$$

$$\hat{\mu} = \frac{\hat{\tau}}{\sum_{i=1}^{n} w_i}$$

Generalized ratio estimator: $r = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i x_i}$, $\hat{\mu}_r = r\mu_x$, $\hat{\tau}_r = r\tau_x$

Regression estimator: $\hat{\mu}_G = \tilde{\mu}_y + B(\mu_x - \tilde{\mu}_x)$

Target rate: $\frac{n'}{n}$

Response rate: $\frac{m}{n}$

Project: Estimating Chinook Salmon Escapement

Purpose: The purpose of this project is to provide statistical support to develop survey and analysis methods to estimate the number Chinook salmon redds of transported anadromous fish by watershed in the upper North Fork Lewis River area upstream of Swift Dam.

Background: Swift Dam on the North Fork Lewis River in south central Washington does not have adequate fish passage infrastructure for anadromous fish, so fish are collected below the dam and trucked above the dam to be released. This project is developed to monitor the use of available habitat above the dam for spawning. Because the fish are trucked above the dam, the number of females available to spawn is known.

The sampling frame is provided in Figure 1 and the number of miles per basin is provided in Table 1.

Scope of Work:

1. **Develop a survey design**. Develop a survey design that incorporates a GRTS sample of points for annual redd surveys by field crews.
2. **Outline analysis approach**. Develop an analysis approach that will facilitate the estimation of the total number of redds and the proportion of spawning females above Swift Dam. State your assumptions.

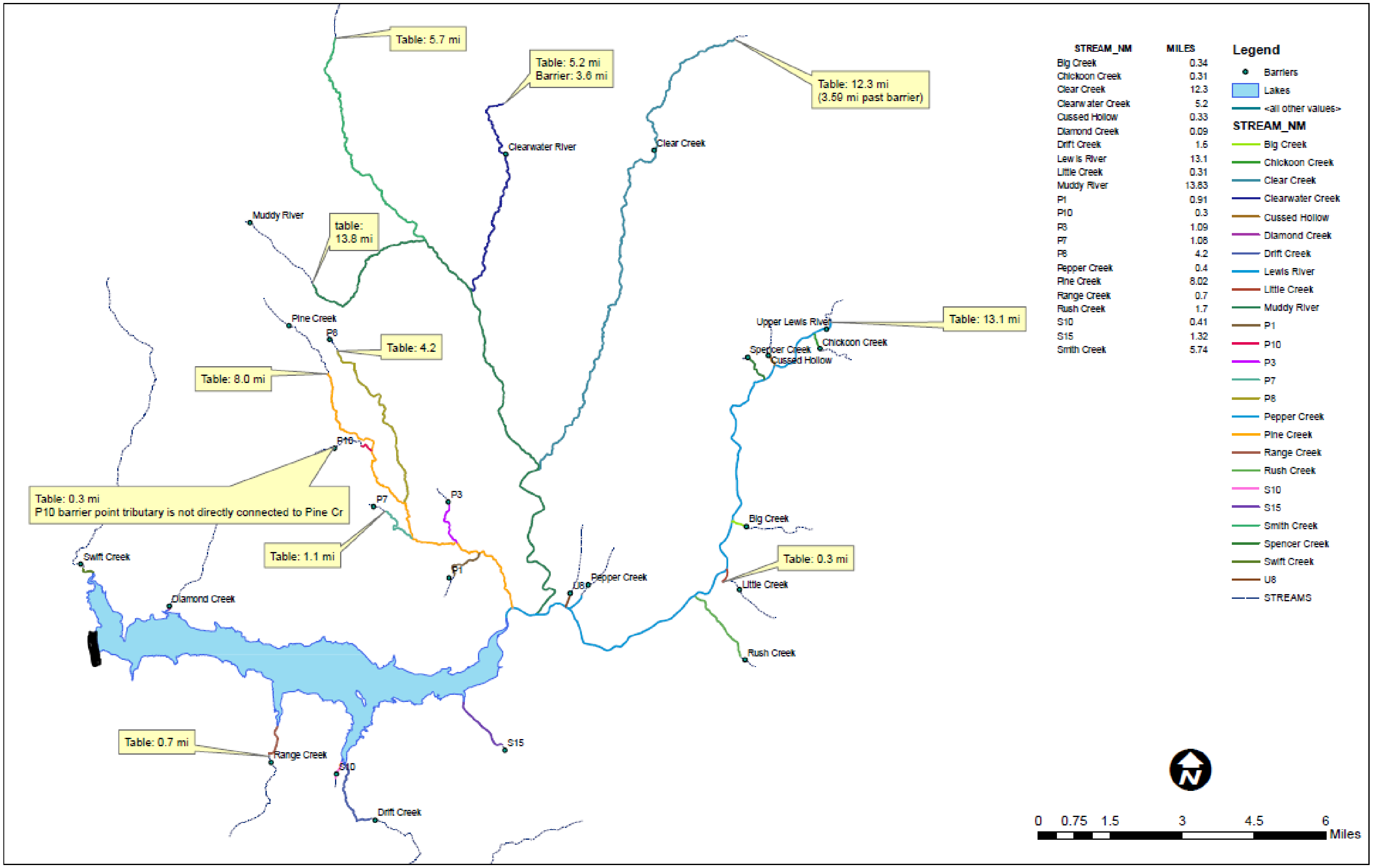

Figure 1: Swift Reservoir, Swift Dam (black bar), and North Fork Lewis, Muddy River, Pine Creek and Swift Creek watersheds.

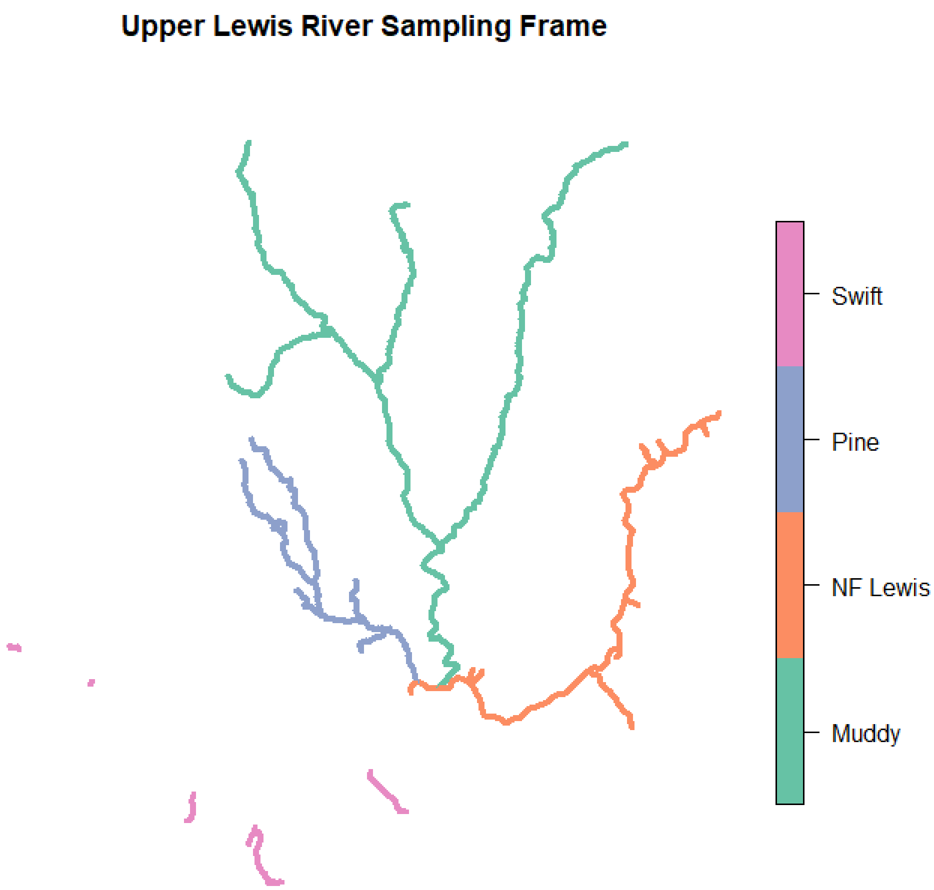

Figure 2: Swift Reservoir, Swift Dam (black bar), and North Fork Lewis, Muddy River, Pine Creek and Swift Creek watersheds.

Table 1: Length of accessible stream by watershed and stream

Watershed	Stream Name	Accessible Reach Length (miles)
NF Lewis River and Minor Tributaries	Mainstem NF Lewis River	13.1
	U8	0.3
	Spencer Creek	0.6
	Pepper Creek	0.4
	Rush Creek	1.7
	Little Creek	0.3
	Big Creek	0.3
	Cussed Hollow	0.3
	Chickoon Creek	0.3
	Total Miles	**17.3**
Muddy River Watershed	Muddy River	13.8
	Clear Creek	12.3
	Clearwater Creek	5.2
	Smith Creek	5.7
	Total Miles	**37.0**
Pine Creek Watershed	Lower Pine Creek	8.0
	P1	0.9
	P3	1.0
	P7	1.1
	P8	4.2
	P10	0.3
	Total Miles	**15.5**
Swift Creek Reservoir Tributaries	Swift Creek	0.3
	Diamond Creek	0.1
	Range Creek	0.7
	S10	0.4
	Drift Creek	1.6
	S15	1.3
	Total Miles	**4.4**
Total Miles in the Sample Universe		**74.2**